SS Tutorial - 3

Q1. Consider a system S with input x[n] and output y[n]. This system is obtained through a series interconnection of a system S1 followed by a system S2. The input-output relationship for S_1 and S_2 are

S_1 : $y_1[n] = 2x_1[n]+4\, x_1\,[n-1]$

S_2 : $y_2[n] = x_2[n-2] + 0.5\, x_2[n-3]$

Where x_1 [n] and x_2[n] denote input signals

(a) Determine the input output relationship for system S.

(b) Does the input-output relationship of system S change if the order in which S_1 and S_2 are connected in series is reversed?

Q2. Consider a discrete time system with input x[n] and output y[n]. The input-output relationship for this system is
y[n] = x[n]x[n-2]

a) Is the system memory-less?

b) Determine the output of the system when input is Aδ[n], where A is any real/complex number
c) Is the system invertible?

Q3. Following system is linear or non-linear, causal or non-causal, time invariant or time varying?

y(t) = x(t). cos(ωt)

Q4. a) Show that if a system is either additive or homogeneous, it has the property that if input is identically zero, then output is also identically zero.

b) Determine a system (either continuous or discrete time) that is *neither* additive *nor* homogeneous but which has a zero output if the input is identically zero.

c) From part a) can you conclude that if input to a linear system is zero between times t1 and t2 in continuous time or between times n1 and n2 in discrete time, then its output must also be zero between these same times? Explain your answer.

Q5. Consider a system S with input x[n] and output y[n] related by

$$y[n]=x[n]\,\{g[n]+g[n-1]\}$$

a) If g[n] = 1 for all n, show that S is time invariant

b) If g[n]= n, show that S is not time invariant

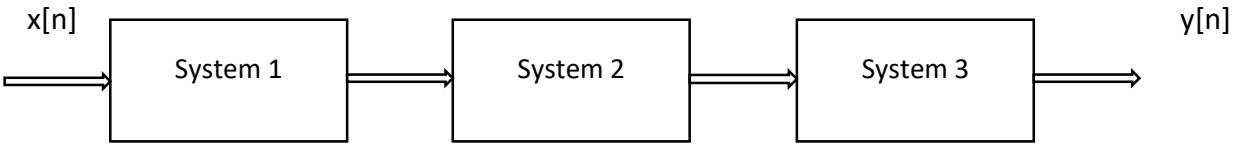
Q6. Are the following statement true or false. Justify your answer.
(a) The series interconnection of two linear time invariant systems is itself a linear, time-invariant system.

(b) The series interconnection of two non linear system is itself non linear.

Q7. Consider three systems with following input output relationships:

System 1: $y[n] = \begin{cases} x[n/2], & n \text{ even} \\ 0 \quad , & n \text{ odd} \end{cases}$

System 2: $y[n] = x[n] + \frac{1}{2}x[n-1] + \frac{1}{4}x[n-2]$

System 3: $y[n] = x[2n]$

Suppose that these systems are connected in series as depicted in figure below. Find the input-output relationship for the overall interconnected system. Is this system linear? Is it time invariant?